

Session 13:

- 1.) In Power BI Desktop, import a sample dataset of IPL cricket matches and create a bar chart showing total runs by each team. Enable drill-down so that clicking on a team reveals runs by individual players from that team.

ANSWERS:

A **Bar Chart** is created in Power BI after importing the IPL cricket dataset to show the **total runs scored by each team**.

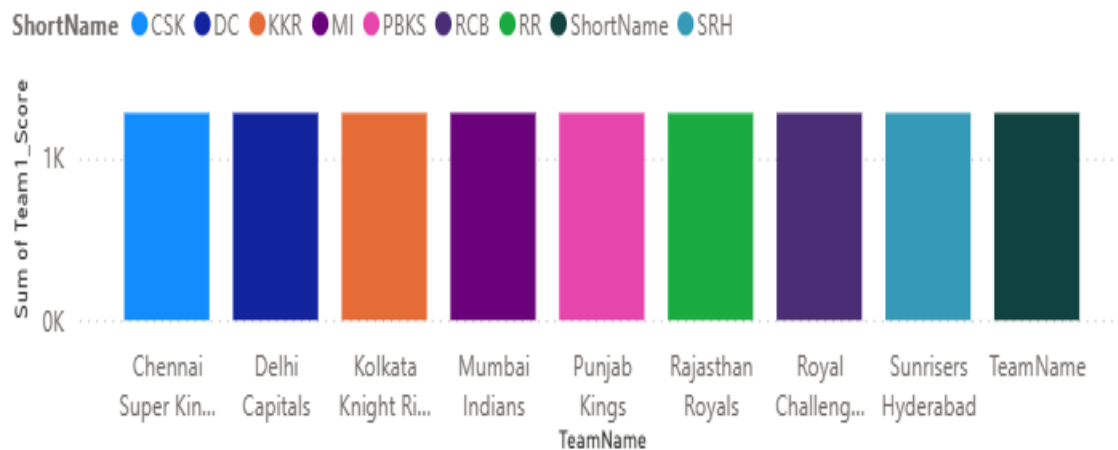
In this visual, the **TeamName** field is placed on the X-axis, and **Team1_Score (Sum)** is used on the Y-axis. This setup helps to compare the total runs scored by different teams such as Chennai Super Kings, Mumbai Indians, Royal Challengers Bangalore, etc.

To enable deeper analysis, a **drill-down feature** is added by creating a hierarchy. The **TeamName** is kept at the top level, and **PlayerName** is added below it in the hierarchy. When the user clicks on a specific team in the bar chart, it drills down to show **runs scored by individual players of that team**.

This makes the chart interactive and more informative, as it first gives an overall team comparison and then allows detailed player-level insights.

Overall, the bar chart clearly shows **team-wise total runs** and, with drill-down, provides a deeper understanding of **player contributions within each team**.

Sum of Team1_Score by TeamName and ShortName



- 2.) Create a hierarchy in Power BI using a dataset of movies (for example: Genre > Year > Movie Title). Build a visual that lets users drill through these levels to see box office collections at each stage.

Hint: Use the 'Manage Hierarchy' option in the Fields pane to set up your levels.

ANSWERS:

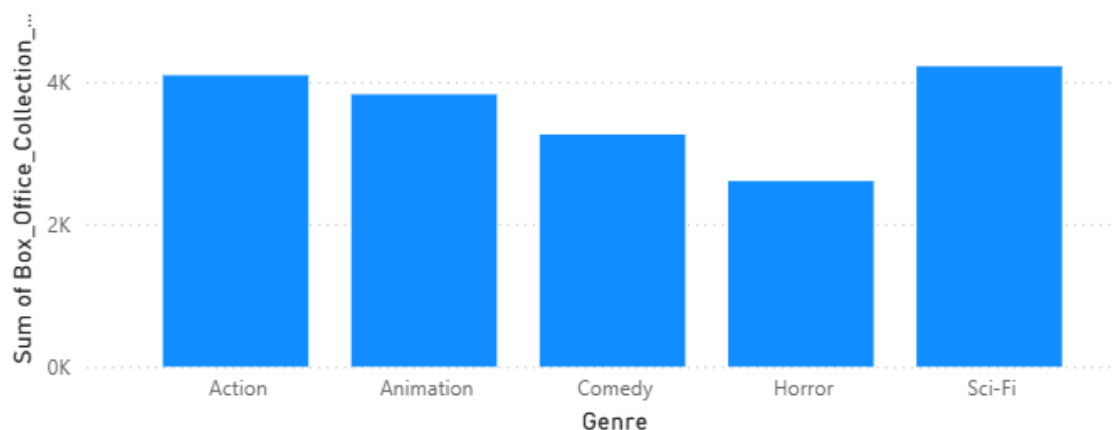
A **hierarchy** is created in Power BI using the dataset of movies by arranging fields in the order **Genre > Year > Movie Title**. This is done using the **Manage Hierarchy** option in the Fields pane, which helps in organizing data in multiple levels.

A visual (such as a **bar chart**) is then created where **Genre** is placed on the X-axis and **Box_Office_Collection_Millions** is used as the value. The hierarchy is added to the Axis so that users can drill down step by step.

When drill-down is enabled, users can first see total collections by **Genre**, then click to view data by **Year**, and further drill down to see collections for each **Movie Title**.

This makes the analysis more interactive and helps to understand box office performance at different levels in a clear and structured way.

Sum of Box_Office_Collection_Millions by Genre



3.) Apply conditional formatting to a table visual displaying restaurant ratings (like Zomato) so that ratings above 4.5 appear in green, between 3 and 4.5 in yellow, and below 3 in red.

ANSWERS:

A **table visual** is created in Power BI to display restaurant data including **Restaurant_Name, Cuisine, and Rating**.

Conditional formatting is applied on the **Rating column** using the **Background Color** option. Rules are set such that:

- Ratings **above 4.5** are shown in **green**,
- Ratings **between 3 and 4.5** are shown in **yellow**,
- Ratings **below 3** are shown in **red**.

This formatting helps to quickly identify **high-performing, average, and low-rated restaurants**. Green highlights the best restaurants, yellow shows moderate performance, and red indicates poor ratings.

Overall, conditional formatting makes the table more **visual, clear, and easy to analyze at a glance**.

Restaurant_Name	Cuisine	Sum of Rating
6 Ballygunge Place	Chinese	4.50
Aminia	Chinese	2.00
Arsalan	Chinese	2.00
Bademiya	Continental	2.60
Barbeque Nation	Desserts	2.80
Bhojohori Manna	Mughlai	3.60
Britannia & Co.	South Indian	3.30
Bukhara	Fast Food	2.00
Burger King	Desserts	2.10
Cafe Coffee Day	Mughlai	4.30
Chaayos	South Indian	2.60
Chili's	Fast Food	4.20
Corner House	Desserts	2.10
Dominos	Chinese	1.70
Farzi Cafe	Desserts	3.90
Flurys	North Indian	1.70
Girish Chandra Dey	Chinese	4.80
Hard Rock Cafe	Chinese	3.30
Indian Accent	Desserts	3.60
Karim's	Desserts	2.50
KFC	Chinese	4.80
Leopold Cafe	North Indian	2.10
McDonald's	Fast Food	1.70
Mocambo	North Indian	3.60
Murugan Idli Shop	Chinese	2.80
Nando's	Chinese	2.20
Oudh 1590	South Indian	4.00
Pa Pa Ya	Fast Food	3.00
Peter Cat	Chinese	4.10
Total		123.50

- 4.) Go to the Power BI Marketplace and add a custom visual of your choice (for example, a 'Word Cloud' or 'Bullet Chart'). Use it to visualize the most frequently ordered food items from a Swiggy-style order dataset.

Hint: After importing, drag the visual onto your report and assign the relevant field.

ANSWERS:

- 5.) Use ChatGPT to help you write a DAX formula that calculates the average delivery time per city from a Flipkart orders dataset, then apply this measure to a visual in your Power BI report.

Hint: Paste your ChatGPT prompt and the generated DAX formula in your submission.

ANSWERS:

A **DAX measure** is created in Power BI to calculate the average delivery time for each city from the Flipkart orders dataset.

The following formula is used:

Average Delivery Time = AVERAGE(Orders[Delivery_Time])

This measure calculates the **mean delivery time** based on the Delivery_Time column.

After creating the measure, a **table visual** is added where:

- **City** is placed in the rows
- **Average Delivery Time (measure)** is placed in the values

This setup displays the **average delivery time for each city**, as shown in the visual.

The result helps to easily compare delivery performance across different cities. Cities with lower average time indicate **faster delivery**, while higher values show **slower delivery performance**.

Overall, this measure makes the analysis simple and helps in identifying which cities are performing better in terms of delivery speed.

City	Average Delivery Time
Ahmedabad	1.62
Bengaluru	2.15
Chennai	2.07
Delhi	1.76
Hyderabad	1.84
Jaipur	1.65
Kolkata	1.54
Mumbai	1.76
Pune	1.78
Surat	1.61
Total	1.79